

1. A memory system is using paging with TLBs. The CPU generates 2400 memory references were performed. Of these memory references 1800 were hits. There were _____ memory accesses to the TLB.

2400

2. TLBs are used to _____.

minimize the memory access time

3. A memory system is using paging with TLBs. The CPU generated 2400 memory references were performed. Of these memory references 1800 were hits. There were _____ memory accesses to the main memory.

3000

4. With classical contiguous allocation (no paging), it takes _____ memory access(es) to access an instruction or data.

1

5. With simple paging, it takes _____ memory access(es) to access an instruction or data.

2

6. A memory system is using paging with TLBs. The CPU generated 2400 memory references were performed. Of these memory references 1800 were hits. The hit ratio is then _____.

0.75

7. A memory system is using paging with TLBs. The CPU generated 5000 memory references were performed. Of these memory references 4500 were hits. We assume that the memory time access to the main memory is 1 (some unit time). The access time to the TLBs is $1/20$ (0.05). We want to find the effective (average) access time for the memory references generated above by the CPU.

1) The hit ratio is _____.

2) In total, the memory references that were hits took _____ access time.

3) The number of misses were _____.

4) In total, the memory references that were misses took _____ access time.

5) The average access time for all memory references is _____.

6) Based on the formula for the effective (average) access given in the lecture, the effective (average) access time for all memory references is _____.

7) The difference between the results of Question 5 and 6 should be _____.

0.90

4725

500

1025

1.15

1.15

0

8. A memory system is using paging with TLBs. The CPU generated 4000 memory references were performed. Of these memory references 3800 were hits. We assume that the memory time access to the main memory is 1 (some unit time). The access time to the TLBs is $1/20$ (0.05). We want to find the effective (average) access time for the memory references generated above by the CPU.

1) The hit ratio is _____.

2) In total, the memory references that were hits took _____ access time.

- 3) The number of misses were _____.
4) In total, the memory references that were misses took _____ access time.
5) The average access time for all memory references is _____.
6) Based on the formula for the effective (average) access given in the lecture, the effective (average) access time for all memory references is _____.
7) The difference between the results of Question 5 and 6 should be _____.

0.95

3990

200

410

1.1

1.1

0